CS201 Lab 8

1. *Modifying the Euler method to better tackle oscillatory problems:* Recall that the Euler method does not conserve energy, which makes it unsuitable for oscillatory problems. In this problem, a slight modification to the Euler algorithm to improve it. To illustrate, consider the oscillatory problem,
$$\ddot{x} = -x. \tag{1}$$
Following Euler's algorithm, the system is discretized as follows,
$$\begin{aligned} v_{n+1} &= v_n - \Delta t x_n, \\ x_{n+1} &= x_n + \Delta t v_n. \end{aligned}$$
For oscillatory problems, a more suitable algorithm may be devised by replacing v_n by v_{n+1} in the second equation above. The improved algorithm, which is known as Euler-Cromer algorithm is given by,
$$\begin{aligned} v_{n+1} &= v_n - \Delta t x_n, \\ x_{n+1} &= x_n + \Delta t v_{n+1}, \end{aligned}$$
i.e., the present velocity is used to update the present position rather than the past velocity.

   Simulate eqn.(1) by both the Euler and Euler-Cromer algorithms. You may choose suitable initial conditions. Display t vs $x(t)$ for three step-sizes for each method so that the accuracies of the algorithms can be compared. Also plot the value of the energy, $E_n = \frac{1}{2}(v_n^2 + x_n^2)$ at each time step (you may do this for one single step-size only).

2. To see why the Euler-Cromer method is better at conserving energy, compute and show that,
$$E_{n+1} - E_n = (v_n^2 - x_n^2)(\Delta t)^2 - 2v_n x_n(\Delta t)^3 + x_n^2(\Delta t)^4.$$
To see why this is an improvement for oscillatory problems, it is best to consider the simple harmonic motion in eqn.(1). Show that the coefficients of $(\Delta t)^2$ and $(\Delta t)^3$ average to zero over one time period (cycle), so that the Euler-Cromer method conserves energy up to $(\Delta t)^4$.

3. The equation of motion of the damped oscillator is given by $\ddot{x} + 2b\dot{x} + \omega_0^2 x = 0$. Solve this differential equation for $x(0) = 0, v(0) = 1$, setting $\omega_0 = 1$, and up to at least three complete cycles. Adjust b suitably and plot t vs $x(t)$ (using Euler-Cromer algorithm) for underdamped, overdamped and critically damped oscillations.